STANDALONE TECHNICAL SUMMARY

Two Kalman-Related Recursions, Two Different Estimation Problems

Fast-MHE arrival-cost covariance versus NeuroMHE's auxiliary Kalman-filter-based gradient algorithm

Verified central distinction. Zavala, Laird, and Biegler (2008), Sec. 3.3, extract covariance information from the *original* MHE KKT/Riccati system for updating the next *arrival cost*. In Sec. 4 they describe a Kalman-like update of the original *state estimate*. **Neither is a Kalman-filter recursion for derivatives of MHE estimates with respect to weighting parameters.** NeuroMHE (2024), by contrast, explicitly constructs an *auxiliary sensitivity-estimation problem* and derives a Kalman-based solver for its matrix-valued gradients. [1–3]

1. Original MHE: state covariance used to update the arrival cost

Let $\Sigma_{\text{arr}}(k)$ be the covariance associated with the original MHE's arrival estimate, and let $\bar{x}_b$ be the arrival center. In the statistical weighting convention, the arrival term is

$$\Gamma_b(x_b) = \frac{1}{2}(x_b - \bar{x}_b)^\top \Sigma_{\text{arr}}(k)^{-1}(x_b - \bar{x}_b). \tag{1}$$

Notation in fast MHE: the paper calls this covariance $P_0(k)$, so its original arrival penalty uses $P_0(k)^{-1}$. This differs from NeuroMHE's notation, where P already denotes an *information/penalty weight* in the objective. [1,2]

Fast MHE Sec. 3.3, Eqs. (21)–(23), extracts matrices Π_0 and M_1 from the original estimator's Riccati/KKT coefficients. After omitting the nonlinear second-order and state–noise cross terms specified there, the formulas take the familiar EKF form:

$$\Pi_0 = \Sigma_{\text{arr}}(k) - \Sigma_{\text{arr}}(k)H_0^\top \left(H_0\Sigma_{\text{arr}}(k)H_0^\top + R_0\right)^{-1} H_0\Sigma_{\text{arr}}(k), \tag{2}$$

$$M_1 = A_0\Pi_0 A_0^\top + G_0 Q_0 G_0^\top. \tag{3}$$

Here H_0 is the output Jacobian (called C_0 in the 2008 paper), and R_0, Q_0 are the original measurement and process noise covariance matrices. The next window is supplied with

$$\boxed{\Sigma_{\text{arr}}(k+1) = M_1, \qquad \text{arrival penalty weight} = M_1^{-1}.} \tag{4}$$

This is the same *forward dynamic-programming / Kalman covariance* principle derived explicitly in the earlier report [3]. In its *Sec. 4.3, Eq. (15), p. 3*, the prediction step is

$$P_{i+1}^{\text{DP}} = F_i C_i^{\text{DP}} F_i^\top + G_i Q_i G_i^\top, \qquad C_i^{\text{DP}} = P_{i|i}. \tag{5}$$

That report's *Sec. 4.3, Eq. (16), p. 6* shows that the resulting forward-DP cost-to-come is

$$\Gamma_{i+1}(z) = \tfrac{1}{2}(z - \hat{x}_{i+1|i})^\top (P_{i+1}^{\text{DP}})^{-1}(z - \hat{x}_{i+1|i}) + \kappa_{i+1}. \tag{6}$$

Its *Sec. 4.4, Eq. (17), p. 7* derives the equivalent one-matrix Riccati covariance recursion; *Sec. 5.1, Eqs. (18)–(19), p. 8* establishes the shifted-window restart using the *inverse* of the propagated covariance as the new arrival weight. The *Sec. 5.2, p. 8* indexing rule avoids counting the first retained measurement twice. Under the report's linear, unconstrained quadratic assumptions and consistent indexing, (2)–(3) reproduce this filtering/prediction update: $\Pi_0 \leftrightarrow C_i^{\text{DP}}$, $M_1 \leftrightarrow P_{i+1}^{\text{DP}}$. This is an original-state *arrival-covariance* relation, not a parameter-gradient recursion [1,3].

Two separate results in the 2008 paper. The full KKT/Riccati covariance extraction may retain second-order information beyond the simplified EKF. Separately, Sec. 4, Eqs. (24)–(25), expresses a *terminal original-state estimate correction* driven by a new measurement innovation. Neither operation computes a weighting-parameter gradient using a Kalman filter. [1]

2. NeuroMHE: the Kalman coefficients belong to an auxiliary *gradient* estimator

Let θ contain the original MHE's tunable weights. To clarify the notation, use P_a for the *arrival information weight* appearing in NeuroMHE's original objective:

$$\begin{aligned} J_1(x, w; \theta) = \tfrac{1}{2}\|x_b - \bar{x}_b(\theta)\|^2_{P_a(\theta)} + \tfrac{1}{2}\sum_{k=b}^{t} \|y_k - h_k(x_k)\|^2_{R_k(\theta)} \\ + \tfrac{1}{2}\sum_{k=b}^{t-1} \|w_k\|^2_{Q_k(\theta)}, \qquad x_{k+1} = f_k(x_k, w_k). \end{aligned} \tag{7}$$

Here $\|v\|^2_M = v^\top M v$. The matrices Q_k, R_k are *objective weights*; they are inverse noise covariances only if the statistical interpretation is imposed. NeuroMHE defines sensitivity-state variables

$$X_k = \frac{\partial \hat{x}_{k|t}}{\partial \theta}, \qquad W_k = \frac{\partial \hat{w}_{k|t}}{\partial \theta}, \qquad \Lambda_k = \frac{\partial \lambda^*_k}{\partial \theta}. \tag{8}$$

It constructs an *auxiliary matrix-valued MHE* whose quadratic stage coefficients are the original *Lagrangian Hessians*, not simply the original sensor/noise weights. Its cost has the structure [2, Eq. (15)]

$$\begin{aligned} J_2(X, W) = \tfrac{1}{2}\operatorname{tr}\left[(X_b - \bar{X}_b)^\top P_a (X_b - \bar{X}_b)\right] \\ + \sum_{k=b}^{t} \left\{\tfrac{1}{2}\operatorname{tr}(X_k^\top \bar{L}_k^{xx} X_k) + \operatorname{tr}[(L_k^{x\theta})^\top X_k]\right\} \\ + \sum_{k=b}^{t-1} \left\{\tfrac{1}{2}\operatorname{tr}(W_k^\top L_k^{ww} W_k) + \operatorname{tr}(W_k^\top L_k^{wx} X_k) + \operatorname{tr}[(L_k^{w\theta})^\top W_k]\right\}, \end{aligned} \tag{9}$$

subject to $X_{k+1} = F_k X_k + G_k W_k$. At the first stage, $\bar{L}_b^{xx} = L_b^{xx} - P_a$, which removes the arrival weight already displayed in the first term. At later stages $\bar{L}_k^{xx} = L_k^{xx}$.

The auxiliary Kalman-form coefficient recursion [2, Eqs. (17)–(18)] uses

$$P_b^{\text{aux}} = P_a^{-1}, \tag{10}$$

$$P_{k+1}^{\text{aux}} = \bar{F}_k C_k^{\text{aux}} \bar{F}_k^\top + G_k (L_k^{ww})^{-1} G_k^\top, \tag{11}$$

$$C_k^{\text{aux}} = \left(I - P_k^{\text{aux}} S_k\right)^{-1} P_k^{\text{aux}}, \qquad S_k = L_k^{xw}(L_k^{ww})^{-1} L_k^{wx} - \bar{L}_k^{xx}. \tag{12}$$

Why the initial inverse looks the same. The *arrival quadratic coefficient* P_a is shared by the original and auxiliary objectives, explaining $P_b^{\text{aux}} = P_a^{-1}$. But the auxiliary variable is $X_b = \partial \hat{x}_b / \partial \theta$, not x_b, and its stage coefficients are Lagrangian Hessians and mixed θ-derivatives. **This initialization is not a new EKF covariance update and does not update the physical MHE arrival cost.**

3. Related algebra does not mean the two covariance calculations are the same

For example, if the original model and measurement equations are nonlinear, the sensitivity quadratic coefficients may include curvature terms such as

$$L_k^{ww} = Q_k - \sum_a \lambda^*_{k,a} \nabla^2_{ww} f_{k,a}, \tag{13}$$

$$L_k^{xw} = -\sum_a \lambda^*_{k,a} \nabla^2_{xw} f_{k,a}, \tag{14}$$

for the dynamics multiplier convention $\lambda_k^\top [x_{k+1} - f_k(x_k, w_k)]$, in addition to measurement-residual curvature in L_k^{xx}. These are not simply conventional EKF noise covariances.

Possible numerical coincidence is not conceptual identity. For a linear-quadratic model (or an approximation neglecting these second-order terms), the auxiliary Hessians can reduce to $L_k^{ww} = Q_k$, $L_k^{xw} = 0$, and $\bar{L}_k^{xx} = H_k^\top R_k H_k$. Under matching initialization, $P_k^{\text{aux}}, C_k^{\text{aux}}$ may then follow exactly the same *numerical covariance equations* as the original estimator. But $\hat{X}_{k|k}^{\text{KF}}$ still denotes a parameter-sensitivity trajectory, not the original state estimate. [2,3]

4. Why the full fast-MHE → NeuroMHE transformation is non-trivial

The common numerical structure can be made explicit. Let Π_k^Z, M_k^Z denote fast MHE's Riccati coefficient matrices (not its extracted EKF quantities as statistical outputs), and let $P_k^{\text{aux}}, C_k^{\text{aux}}$ denote NeuroMHE's auxiliary filter coefficients. Under matching differential-KKT Hessian blocks and conditions, the matrix recursion admits the algebraic correspondence

$$\boxed{\Pi_k^Z = C_k^{\text{aux}}, \qquad M_k^Z = P_k^{\text{aux}}.} \tag{15}$$

For example, with $D_k = L_k^{xw}$, $R_k^H = L_k^{ww}$, and $\bar{F}_k = F_k - G_k(R_k^H)^{-1}D_k^\top$, both methods have the same stage Schur-complement prediction:

$$\underbrace{M_{k+1}^Z}_{\text{fast MHE}} = \bar{F}_k \underbrace{\Pi_k^Z}_{\text{fast MHE}} \bar{F}_k^\top + G_k(R_k^H)^{-1}G_k^\top = \underbrace{P_{k+1}^{\text{aux}}}_{\text{NeuroMHE}}. \tag{16}$$

This *coefficient-level* transformation is comparatively direct after aligning notation and applying standard Schur-complement identities. **It is not the complete Kalman gradient algorithm.**

1. **The linear-system right-hand side must change.** Fast MHE Eqs. (12)–(14) are presented for the original MHE Newton/search-direction system; sensitivity requires the *parameter derivatives of the KKT equations*, as in its general sensitivity Eq. (16). These are not simply Newton residuals.

2. **Affine recursions must be transformed.** Fast MHE's residual-based $r_{M,k}$ must be reinterpreted through an appropriate change of variables as the negative predicted sensitivity $-\hat{X}_{k|k-1}^{\text{KF}}$ (for aligned interior stages and parameter-independent dynamics). The filtered sensitivity $\hat{X}_{k|k}^{\text{KF}}$ and its parameter-gradient forcing terms do not appear explicitly in fast MHE.

3. **Inherited arrival-center sensitivity must be included.** The initial sensitivity equations contain $\bar{X}_b = \partial \bar{x}_b / \partial \theta$ and, when the weight depends on θ, additional mixed-derivative terms. Matching only the coefficient recursion misses this forcing. NeuroMHE treats this explicitly, with a past-window approximation in its Appendix A.

4. **Backward smoothing and the boundary convention must be recovered.** NeuroMHE establishes

$$\hat{X}_{k|t} = \hat{X}_{k|k}^{\text{KF}} + C_k^{\text{aux}} \bar{F}_k^\top \Lambda_k^*, \qquad \Lambda_t^* = 0 \;\Rightarrow\; \hat{X}_{t|t} = \hat{X}_{t|t}^{\text{KF}}. \tag{17}$$

   Obtaining this interpretation requires aligning multiplier indices and transforming the full recovery equations, not just their forward matrix coefficients.

5. **The auxiliary optimum must be proved to represent the gradients.** NeuroMHE Lemma 1 identifies the differential KKT equations with the KKT conditions of (9); Lemma 2 derives the Kalman-form solution. This *optimization-level correspondence* does not follow merely from writing (15).

5. Final assessment: the precise contribution and its limits

Final assessment. The conversion from the fast-MHE Riccati/Newton equations to NeuroMHE's complete Kalman-filter-based *parameter-gradient* solver is non-trivial. The underlying Schur-complement matrix operations are related and can be matched under identical KKT coefficients, right-hand sides, and invertible stage operations. Yet the complete sensitivity algorithm additionally requires a changed right-hand side, affine gradient propagation, inherited-prior derivatives, dual-variable smoothing, and proof that the auxiliary MHE optimal trajectories are the parameter sensitivities. These additional ingredients are explicitly developed in DMHE/NeuroMHE and are not given as a Kalman *gradient* filter in fast MHE (2008). This supports a **substantive structural and methodological contribution relative to the cited fast-MHE paper**, without asserting a newly invented Riccati elimination identity or the first-ever $O(N)$ sensitivity solve.

Three statements that should never be conflated

1. Fast MHE **Sec. 3.3**: extraction of original-state covariance and an EKF-like *arrival-covariance* update; after computing M_1, the next original arrival penalty uses M_1^{-1}.
2. Fast MHE **Sec. 4**: a Kalman-like *measurement-innovation correction* of the original terminal state estimate.
3. NeuroMHE **Eqs. (15)–(20)**: an explicit auxiliary *parameter-sensitivity estimation* problem, solved by its own Kalman-form matrix and sensitivity recursions.

The shared arrival coefficient explains the similar initialization but not the identity of the algorithms. In NeuroMHE the original arrival *information weight* P_a and auxiliary arrival *quadratic coefficient* are the same. Its inverse initializes the auxiliary filter. Only if P_a was itself chosen as the inverse original state covariance will P_a^{-1} numerically equal that covariance at the boundary. The subsequent gradient-filter matrices and the quantities computed from them still belong to the auxiliary sensitivity calculation.

Source notes

[1] V. M. Zavala, C. D. Laird, and L. T. Biegler, "A fast moving horizon estimation algorithm based on nonlinear programming sensitivity," *Journal of Process Control*, 18 (2008), 876–884. Critical locations: Sec. 3.1, Eqs. (9), (12)–(14); Sec. 3.2, Eq. (16); Sec. 3.3, Eqs. (21)–(23); Sec. 4, Eqs. (24)–(25).

[2] B. Wang, Z. Ma, S. Lai, and L. Zhao, "Neural Moving Horizon Estimation for Robust Flight Control," *IEEE Transactions on Robotics*, 40 (2024), 639–659. Critical locations: original MHE Eq. (4); differential KKT Eq. (13); auxiliary MHE Eq. (15); Lemmas 1–2, Eqs. (17)–(20), and Appendices A–C.

[3] "Moving Horizon Estimation, Forward DP, and Riccati Arrival Costs," user-supplied 14-page technical report (19 October 2026), file `MHE_forward_DP_Riccati_arrival_cost_summary_algebra_verified.pdf`. Explicit cross-references: Sec. 2, Eq. (6) (forward Bellman step, p. 1); Sec. 4.3, Eq. (15) (predicted covariance, p. 3), Eq. (16) (arrival function, p. 6); Sec. 4.4, Eq. (17) (covariance Riccati update, p. 7); Sec. 5.1, Eqs. (18)–(19) (new-window initialization, p. 8); Sec. 5.2 (avoiding duplicate measurement information, p. 8). Its Sec. 10.3 treats the *distinct* auxiliary gradient solver.

[4] B. Wang *et al.*, "Differentiable Moving Horizon Estimation for Robust Flight Control," *IEEE Conference on Decision and Control*, 2021. Precursor of the auxiliary matrix-valued sensitivity-MHE construction and Kalman-based gradient recursion.

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Scope. This note summarizes and checks the specific claims against the cited supplied sources. The equation-level mapping is a retrospective algebraic analysis, not an attribution of a Kalman-based gradient derivation to the 2008 authors.